

Solving the Two-Market Model

Pascal Michailat
<https://pascalmichailat.org/w/>

Solve the two-market model (x, θ) such that

$$\begin{cases} y^d(x) = y^s(x, \theta) & (AD = AS) \\ l^d(x, \theta) = l^s(\theta) & (LD = LS) \end{cases}$$

- $y^d(x) = y^s(x, \theta)$

$$\Rightarrow \frac{x^\xi}{[1 + \tau(x)]^{\xi-1}} \cdot \frac{w}{p} = f(x) \cdot a \cdot \left[\frac{\hat{f}(\theta) l}{1 + \hat{\tau}(\theta)} \right]^\alpha$$

$$\Rightarrow f(x) [1 + \tau(x)]^{\xi-1} = \frac{x^\xi w}{p a l^\alpha} \cdot \left[\frac{1 + \hat{\tau}(\theta)}{\hat{f}(\theta)} \right]^\alpha \quad (P)$$

- $l^d(x, \theta) = l^s(\theta)$

$$\Rightarrow \left[\frac{f(x) a \alpha}{w/p} \right]^{1/(1-\alpha)} \left[\frac{1}{1 + \hat{\tau}(\theta)} \right]^{\alpha/(1-\alpha)} = \hat{f}(\theta) l$$

$$\Rightarrow f(x) = \frac{w/p \cdot l^{1-\alpha}}{a \alpha} \hat{f}(\theta)^{1-\alpha} [1 + \hat{\tau}(\theta)]^\alpha \quad (L)$$

What do we learn from (L)

$$x = x^L(\theta) = f^{-1} \left(\underbrace{\frac{w/p}{a \alpha} l^{1-\alpha} \hat{f}(\theta)^{1-\alpha}}_{\text{at } \theta=1} \underbrace{[1 + \hat{\tau}(\theta)]^\alpha}_{\text{at } \theta=1} \right)$$

- x^L is strictly $\uparrow$ in θ

- $x^L(0) = f^{-1}(0) = 0$

- θ^L st $x^L(\theta^L) = f^{-1}(1) = +\infty$
with $\theta^L < \theta^m$

$$I. \left(\frac{x^\varepsilon \mu d}{w \cdot h} \right)^{1/k-1} - 1 > 0$$

↖ ↗ $\lambda > 1$

Then $\lim_{\theta \rightarrow \infty} x^P(\theta) = x^P$ at $\theta \rightarrow \infty$

$$\tau(x^P) = \lambda^{1/k-1} \text{ or } x^P = \tau^{-1}(\lambda^{1/k-1})$$

II. $\lambda < 1$. There is θ^P such that

$$\left(\frac{x^\varepsilon \mu d}{w \cdot h} \cdot \frac{1}{f'(\theta^P)} \right)^{1/k-1} = 1$$

$$\Leftrightarrow f'(\theta^P) = \frac{x^\varepsilon \mu d}{w \cdot h} \Rightarrow \lambda \leftarrow \frac{x^\varepsilon \mu d}{w \cdot h}$$

$$\Leftrightarrow \theta^P = f^{-1}(\lambda)$$

$$\text{Then } x^P(\theta^P) = \tau^{-1}(0) = 0$$